
MS250 Datasheet

Overview

The Cisco Meraki MS250 series switches provide reliable access switching ideal for deploying in branches and small campuses. With stacking capabilities and 10G SFP+ uplinks on every model, redundancy and performance are guaranteed. This family also supports redundant, field-replaceable power supplies for mission-critical networks.

With 5 different models, capable of providing up to 740W of PoE+ power over a variety of port densities, including SFP+ capable uplinks, the MS250 line is fully ready to support branch and small campus infrastructure deployments.



Features

- Managed via Cisco Meraki Dashboard
- Remote Packet Capture Tools via Meraki Dashboard
- Automatic Firmware upgrades
- SNMP/Syslog Integration
- IPv4/6 ACL support
- 802.1q VLAN tagging
- L3 Switching including OSPF
- 2x Dedicated Stack Ports providing 80G of Stacking bandwidth
- Broadcast Storm Control
- Dynamic ARP Inspection / DHCP Snooping
- 802.1X Authentication
- 10/100/1000 Mbps RJ45
- 4x 1000/10000 Mbps SFP+
- PoE+ models available for device level powering
- Warm Spare capable

Configuration

The basic initial configuration of the MS250 is just as simple as any other model of MS switch. The links below provide additional information and instructions relating to each step in getting the device setup and configured for the first time.

1. [Claim the device to an Organization on the Meraki Dashboard](#)
 - a. If a Dashboard Organization does not yet exist, [Create one](#)
2. [Add the device to a Dashboard Network](#)
 - a. If a Network does not yet exist, [Create one first](#)
3. Physically connect the device to the local network
 - a. Connect one of the RJ45 ports to existing infrastructure to provide a temporary uplink
 - b. Power on the device and let it check in to the Dashboard
 - c. If necessary, configure a Static IP through the [Local Status Page](#) to allow it to communicate with the Meraki Dashboard.
4. Allow the device to complete check-in and perform any initial firmware upgrades
5. Finish configuring the device from the Meraki Dashboard
 - a. [Create a Switch Stack](#)
 - b. [Manage local VLANs / Port configuration](#)
 - c. [Configure Layer 3 Routing](#)

Context and Comparisons

	MS225-48FP	MS250-48LP	MS250-48FP
1GbE RJ45	48	48	48
10GbE SFP+	4	4	4
Hardware Stack Port	2	2	2
Dedicated Mgmt Interface	1	1	1
Hot Swap Power Supply	-	Yes, Dual	Yes, Dual
Layer 3 Routing	Static Routing	Yes, OSPF	Yes, OSPF
Max Stacking Bandwidth	80 Gbps	80 Gbps	80 Gbps
Max Switching Capacity	176 Gbps	176 Gbps	176 Gbps

Technical Breakdown

Hardware Breakdown

	MS250-24	MS250-24P	MS250-48	MS250-48LP	MS250-48FP
1GbE RJ45	24	24	48	48	48
10GbE SFP+	4	4	4	4	4
Hardware Stack Port	2	2	2	2	2
Dedicated Mgmt Interface	1	1	1	1	1
Fan Operation	Fixed Internal	Fixed Internal	Fixed Internal	Fixed Internal	Fixed Internal
Hot Swap Power Supply	Yes, Dual	Yes, Dual	Yes, Dual	Yes, Dual	Yes, Dual
Optional Power Supply	2 x 250W	2 x 640W	2 x 250W	2 x 640W	2 x 1025W



More information on power specifications can be found in the [MS250 Series Installation Guide](#).

Throughput and Capabilities

	MS250-24	MS250-24P	MS250-48	MS250-48LP	MS250-48FP
PoE/PoE+ Capable	-	Yes, 370W	-	Yes, 370W	Yes, 740W
Layer 3 Routing	Yes	Yes	Yes	Yes	Yes
Switching Capacity	128 Gbps	128 Gbps	176 Gbps	176 Gbps	176 Gbps
Stacking Bandwidth	80 Gbps	80 Gbps	80 Gbps	80 Gbps	80 Gbps

Physical

	MS250-24	MS250-24P	MS250-48	MS250-48LP	MS250-48FP
Mount Type	1U Rack Mount	1U Rack Mount	1U Rack Mount	1U Rack Mount	1U Rack Mount
Dimensions (h x w x d)	1.72 x 19 x 14.89in (4.38 x 48.46 x 37.8cm)	1.72 x 19 x 14.89in (4.38 x 48.46 x 37.8cm)	1.72 x 19 x 18.82in (4.38 x 48.46 x 47.8cm)	1.72 x 19 x 18.82in (4.38 x 48.46 x 47.8cm)	1.72 x 19 x 20.42in (4.38 x 48.46 x 51.87cm)
Weight	9.9 lb (4.5 kg)	10.59 lb (4.8 kg)	11.56 lb (5.24 kg)	12.37 lb (5.61 kg)	12.83 lb (5.82 kg)
Power Load (idle/ max)	14 / 27 W	15 / 434 W	24 / 44 W	25 / 480 W	25 / 874 W

Operating Temperature	23 °F - 122 °F -5°C - 50°C	23 °F - 122 °F -5°C - 50°C	23 °F - 122 °F -5°C - 50°C	23 °F - 122 °F -5°C - 50°C	23 °F - 122 °F -5°C - 50°C
Humidity	5% to 95%	5% to 95%	5% to 95%	5% to 95%	5% to 95%

Troubleshooting

The MS uses LEDs to inform the user of the device's status. When the device powers on, all the Internet LEDs flash twice. Additional functions are described below, from left to right.

Front Panel Components

Item	Function	LED Status	Meaning
1	Power	Solid orange	Switch is unable to connect to the Meraki cloud
		Flashing white	Firmware upgrade in process
		Solid white	Switch is fully operational and connected to the Meraki cloud
		Off	Switch does not have power
2	Restore	N/A	Restore button to clear switch IP and local configuration settings
3	Switch Ports	Off	No link is detected on this port
		Solid orange	10/100 Mbps (1 Gbps on SFP+)
		Solid green	1 Gbps (10 Gbps on SFP+)

In addition, there is a RESTORE button available on the front panel.

Insert a paperclip if a restore is required.

- A brief, momentary press: To delete a downloaded configuration and reboot.
- Press and hold for more than 10 sec: To force the unit into a full factory restore.

Back Panel Components

Item	Function	LED Status	Meaning
1	Management Interface	Green	Connected, used

2	Stack Ports	Green	QSFP stacking ca
		Off	No link is detecte
3	Redundant Power Supplies	Green	Active and functio

Power cords may be ordered separately.



Equipment is to be used only in a restricted access location and installed/operated only by trained service personnel.

Common Troubleshooting

My device is connected to the network but not checking in to the Meraki cloud or shows a solid Orange LED.

Confirm that the device is powered on and has a valid IP address that is able to access the Internet. Use the Local Status Page to get more information about the connectivity status of the device such as if it can successfully reach the Local Gateway, Internet, and/or Meraki Cloud servers. If necessary, contact Meraki Support for additional assistance.

My Status LED is blinking WHITE

A blinking WHITE Status LED indicates that the device is in contact with the Dashboard Cloud servers and is performing a firmware update. This can sometimes take 20-45 minutes or more to complete depending on hardware and other factors.

My Status LED is blinking ORANGE

The device is not able to successfully communicate with the Dashboard Cloud servers or there may be a hardware issue with the device. Check the Local Status Page of the device to confirm the status and reach out to Meraki Support for further troubleshooting.

Event Log

The most common Event Log messages and their meaning are listed below.

Port STP change

Indicates the STP state of the port has changed, lists the relevant port number, previous, and new states. Typically accompanied by a 'Port status change' event.

Port status change

Indicates the link state of the port has changed, lists the relevant port number, old, and new state. Always accompanied by a 'Port STP change' event.

SFP module inserted/removed

Indicates that an SFP module was either inserted or removed, includes SFP module information for inserted events and always lists the relevant port number.

Common Stacking Alerts

View our dedicated [Switch Stacking document](#) for more detailed information about configuring a Switch Stack and common issues.

Ensure all stack members are configured on dashboard, online and connected via their stacking ports.



Note: If connected and configured correctly, the alert will disappear within up to 1 hour. If the error persists, please contact Cisco Meraki Technical Support for further troubleshooting.

This switch's current stack members differ from the dashboard configuration/ Misconfigured Switch.

**This switch's current stack members
differ from the dashboard configuration.**



Misconfigured switch.

This can occur in the following scenarios:

- Stack members are configured on dashboard, but not all members are connected via their stacking ports.
- A stack member has failed or is powered off.

This switch is not connected to a stack/Switch not connected to stack.

**This switch is not connected to a
stack.**



**Switch not connected to
stack.**

This can occur in the following scenarios:

- The switch is configured on dashboard as a stack member, but is not connected to a stack.

This switch does not have a stack configuration/Unconfigured Switch.

**This switch does not have a stack
configuration.**



Unconfigured switch.

This can occur in the following scenarios:

- The switch is physically connected as a stack, but not configured on dashboard as a stack member.